Exercise 1a

The full hamiltonian for the system is given by

$$\mathcal{H} = \mathcal{H}_{\text{atom}} + \mathcal{H}_{\text{int}} + \mathcal{H}_{\text{field}}$$

Let $|i\rangle\langle j| = |i\rangle \otimes \langle j_k|$ be a basis for the system with $|i\rangle$ being eigenstates of the atom and $|j_k\rangle$ representing a field with j photons in mode k. Setting $|1\rangle$ to be zero energy, the atomic hamiltonian is given by

$$\mathcal{H}_{\text{atom}} = \hbar\omega_{12}|2\rangle\langle 2| + \hbar\omega_{13}|3\rangle\langle 3|$$

The radiation field has hamiltonian

$$\mathcal{H}_{\text{field}} = \sum_{k,\lambda} \hbar\nu_{k,\lambda}\left(a^\dagger_{k\lambda}a_{k\lambda} + \frac{1}{2}\right)$$

and the interaction hamiltonian is given by

$$\mathcal{H}_{\text{int}} = -\hat{d}\cdot\hat{E} = -\sum_{\alpha,\alpha'}|\alpha\rangle\langle\alpha|\hat{d}|\alpha'\rangle\langle\alpha'| \cdot \sum_{k,\lambda}\sqrt{\frac{\hbar\nu_k}{2\varepsilon_0 V}}\left(\hat{a}_{k\lambda}\hat{\varepsilon}_{k\lambda}e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger_{k\lambda}\varepsilon^*_{k\lambda}e^{-i\vec{k}\cdot\vec{r}}\right)$$

By assumption, $\langle 2|\hat{d}|1\rangle = \langle 3|\hat{d}|1\rangle = \mu\hat{z}$. Letting $g_k = \frac{\mu}{\hbar}\sqrt{\hbar\nu_k/2\varepsilon_0 V}(\hat{\varepsilon}_k\cdot\hat{z})e^{i\vec{k}\cdot\vec{r}}$, we get

$$\mathcal{H}_{\text{int}} = -\hbar\sum_{k,\lambda} g_k\left(|2\rangle\langle 1| + |3\rangle\langle 1|\right)\hat{a}_{k\lambda} + g^*_k\left(|1\rangle\langle 2| + |1\rangle\langle 3|\right)\hat{a}^\dagger_{k\lambda}$$

To study spontaneous emission, we consider states $|2,0\rangle, |3,0\rangle$ (atom in excited state, no photons) and $|1,1_k\rangle$ (atom in ground state after emitting photon in mode k). We evaluate the matrix elements as

$$\begin{aligned}
\langle 2,0|\mathcal{H}|2,0\rangle &= \hbar\omega_{12}\\
\langle 3,0|\mathcal{H}|3,0\rangle &= \hbar\omega_{13}\\
\langle 1,1_k|\mathcal{H}|1,1_k\rangle &= \hbar\nu_k\\
\langle 2,0|\mathcal{H}|1,1_k\rangle &= -\hbar g_k\\
\langle 3,0|\mathcal{H}|1,1_k\rangle &= -\hbar g_k
\end{aligned}$$

where we ignored the $\hbar\nu_k/2$ term in the field hamiltonian since it adds a constant to the total energy, irrelevant to dynamics. Thus, we get

$$\mathcal{H} = \underbrace{\hbar\omega_{12}|2\rangle\langle 2| + \hbar\omega_{13}|3\rangle\langle 3| + \sum_k \hbar\nu_k|k\rangle\langle k|}_{:=\mathcal{H}_0} - \sum_k(\hbar g_k|2\rangle\langle k| + \hbar g_k|3\rangle\langle k| + \text{h.c.})$$

where we dropped the sum over λ by absorbing it implicitly into the sum over k. Let $\mathcal{H}_1$ be the interaction part of the hamiltonian, i.e $\mathcal{H}_1 = \mathcal{H} - \mathcal{H}_0$. We move to the interaction picture w.r.t $\mathcal{H}_0$ by defining $|\psi_I\rangle = \hat{U}^{-1}|\psi\rangle$ where $\hat{U} = e^{-i\mathcal{H}_0 t/\hbar}$. Then,

$$\begin{aligned} i\hbar\frac{d|\psi_I\rangle}{dt} &= i\hbar\frac{d\hat{U}^{-1}}{dt}|\psi\rangle + i\hbar\hat{U}^{-1}\frac{d|\psi\rangle}{dt} \\ &= -\mathcal{H}_0\hat{U}^{-1}|\psi\rangle + \hat{U}^{-1}(\mathcal{H}_0 + \mathcal{H}_1)|\psi\rangle \\ &= \hat{U}^{-1}\mathcal{H}_1|\psi\rangle \\ &= \hat{U}^{-1}\mathcal{H}_1\hat{U}|\psi_I\rangle \end{aligned}$$

where we used the fact that $\hat{U}^{-1}$ commutes with $\mathcal{H}_0$ in the last step. Thus, our effective hamiltonian in the interaction picture is $\mathcal{H}_{\text{eff}} = \hat{U}^{-1}\mathcal{H}_1\hat{U}$. We note that since $\mathcal{H}_0$ is diagonal, we have

$$\hat{U}^{-1} = e^{i\mathcal{H}_0 t/\hbar} = e^{i\omega_{12}t}|2\rangle\langle 2| + e^{i\omega_{13}t}|3\rangle\langle 3| + \sum_k e^{i\nu_k t}|k\rangle\langle k|$$

Hence, we get

$$\mathcal{H}_{\text{eff}} = -\hbar\sum_k g_k|2\rangle\langle k|e^{-i(\nu_k-\omega_{12})t} + g_k|3\rangle\langle k|e^{-i(\nu_k-\omega_{13})t} + \text{h.c.}$$

Exercise 1b

Let $|\psi_I(t)\rangle = c_2(t)|2\rangle + c_3(t)|3\rangle + \sum_k c_k(t)|k\rangle$. We want to remove the explicit time dependence in $\mathcal{H}_{\text{eff}}$. First, we bring up the energy of the $|k\rangle$ by making the unitary transformation $|\overline{\psi}\rangle = U_1|\psi\rangle$, with $U_1 = e^{-i\sum_k(\nu_k-\omega_{13})t|k\rangle\langle k|} = \sum_k e^{-i(\nu_k-\omega_{13})t}|k\rangle\langle k|$. The last step holds since $|k\rangle$ is a basis element so $\sum_k|k\rangle\langle k|$ is diagonal. We see that

$$i\hbar\frac{d|\overline{\psi}\rangle}{dt} = i\hbar\left(\frac{dU_1}{dt}|\psi\rangle + U_1\frac{d|\psi\rangle}{dt}\right) = \underbrace{(i\hbar\frac{dU_1}{dt}U_1^\dagger + U_1\mathcal{H}_{\text{eff}}U_1^\dagger)}_{:=\overline{\mathcal{H}}_{\text{eff}}}|\overline{\psi}\rangle$$

We have

$$i\hbar\frac{dU_1}{dt}U_1^\dagger = i\hbar\left(\frac{d}{dt}e^{-i\sum_k(\nu_k-\omega_{13})t|k\rangle\langle k|}\right)U_1 = \hbar\sum_k(\nu_k-\omega_{13})|k\rangle\langle k|$$

and

$$U_1\mathcal{H}_{\text{eff}}U_1^\dagger = -\hbar\sum_k g_k|2\rangle\langle k|e^{-i\omega_{23}t} + g_k|3\rangle\langle k| + \text{h.c.}$$

where $\omega_{23} = \omega_{13} - \omega_{12}$. Hence,

$$\overline{\mathcal{H}}_{\text{eff}} = \hbar\sum_k(\nu_k-\omega_{13})|k\rangle\langle k| - \hbar\sum_k g_k|2\rangle\langle k|e^{-i\omega_{23}t} + g_k|3\rangle\langle k| + \text{h.c.}$$

To get rid of the $e^{-i\omega_{23}t}$ dependence, we bring down state $|2\rangle$ by ω_{23} with the unitary transform $|\widetilde{\psi}\rangle = U_2|\overline{\psi}\rangle$ with $U_2 = e^{i\omega_{23}t|2\rangle\langle 2|} = e^{i\omega_{23}t}|2\rangle\langle 2|$ since $|2\rangle$ is a basis element. Then, similar as before, we find

$$\widetilde{\mathcal{H}}_{\text{eff}} = -\hbar\omega_{23}|2\rangle\langle 2| + \hbar\sum_k(\nu_k - \omega_{13})|k\rangle\langle k| - \hbar\sum_k g_k|2\rangle\langle k| + g_k|3\rangle\langle k| + \text{h.c.}$$

Now, we look at the schrodinger equation, $i\hbar\frac{d|\widetilde{\psi}\rangle}{dt} = \widetilde{\mathcal{H}}_{\text{eff}}$, where $|\widetilde{\psi}\rangle = \tilde{c}_2(t)|2\rangle + \tilde{c}_3(t)|3\rangle + \sum_k \tilde{c}_k(t)|k\rangle$, to get

$$\begin{aligned}
\dot{\tilde{c}}_2 &= i\omega_{23}\tilde{c}_2 + i\sum_k g_k\tilde{c}_k \\
\dot{\tilde{c}}_3 &= i\sum_k g_k\tilde{c}_k \\
\dot{\tilde{c}}_k &= i\underbrace{(\omega_{13} - \nu_k)}_{:=\delta_k}\tilde{c}_k + ig_k^*\tilde{c}_2 + ig_k^*\tilde{c}_3
\end{aligned}$$

We integrate $\dot{\tilde{c}}_k$ using the integrating factor $e^{-i\delta_k t}$ to get

$$c_k(t) = ig_k^*\int_0^t e^{-i\delta_k(t'-t)}(\tilde{c}_2(t') + \tilde{c}_3(t'))dt'$$

where we imposed the intial condition $\tilde{c}_k(0) = 0$. We assume $\tilde{c}_2$ and $\tilde{c}_3$ vary slowly so that

$$c_k(t) = ig_k^*(\tilde{c}_2(t) + \tilde{c}_3(t))\int_0^t e^{-i\delta_k(t'-t)}dt' = ig_k^*(\tilde{c}_2(t) + \tilde{c}_3(t))(\pi\delta(\delta_k) + i\mathcal{P}\frac{1}{\delta_k})$$

Plugging this into $\tilde{c}_2$ gives us

$$\dot{\tilde{c}}_2 = i\omega_{23}\tilde{c}_2 - (\tilde{c}_2 + \tilde{c}_3)\sum_k |g_k|^2(\pi\delta(\delta_k) + i\mathcal{P}\frac{1}{\delta_k})$$

The $i\mathcal{P}(1/\delta_k)$ term leads to energy shifts (lamb shifts in this context) which we can ignore. Invoking results form the notes, we evaluate

$$\sum_k |g_k|^2\pi\delta(\delta_k) = \frac{\mu^2\omega_{13}^2}{3\pi\varepsilon_0\hbar c^3} \approx \frac{\mu^2\omega_{12}^2}{3\pi\varepsilon_0\hbar c^3} := \frac{\Gamma}{2}$$

where was assume $\omega_1 3 \approx \omega_1 2$. All together, we get

$$\dot{\tilde{c}}_2 = i\omega_{23}\tilde{c}_2 - (\tilde{c}_2 + \tilde{c}_3)\Gamma/2 = (i\omega_{23} - \Gamma/2)\tilde{c}_2 - (\Gamma/2)\tilde{c}_3$$

For $\dot{\tilde{c}}_3$, we similarly get

$$\begin{aligned}
\dot{\tilde{c}}_3 &= -(\tilde{c}_2 + \tilde{c}_3)\sum_k |g_k|^2(\pi\delta(\delta_k) + i\mathcal{P}\frac{1}{\delta_k}) \\
&= -(\Gamma/2)\tilde{c}_3 - (\Gamma/2)\tilde{c}_2
\end{aligned}$$

Exercise 1c